

Optimization-based machine learning for computational imaging

Bi-level optimization



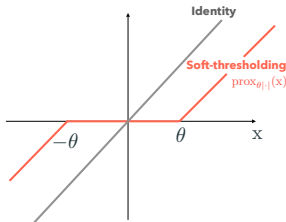
Inversion models: Focus on variational approaches

Minimization problem: $\hat{x}_\theta \in \underset{x}{\operatorname{Argmin}} f_\theta(x) + g_\theta(x)$

Iterative scheme: e.g. **Forward backward algorithm:**

$$\begin{aligned} (\forall k \in \mathbb{N}) \quad x_{k+1} &= \varphi_{k+1}(x_k, \theta) \\ &= \operatorname{prox}_{\tau_k g_\theta}(x_k - \tau_k \nabla f_\theta(x_k)), \end{aligned}$$

Key tool – Proximity operator



$$\begin{aligned} \operatorname{prox}_g(x) &= \arg \min_u \frac{1}{2} \|x - u\|^2 + g(u) \\ &= (\mathbf{I} + \partial g)^{-1} x \end{aligned}$$

[Prox repository]

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$$\begin{aligned} (\forall k \in \mathbb{N}) \quad x_{k+1} &= \varphi_{k+1}(x_k, \theta) = \mathbf{T}(x_k) \\ &= \operatorname{prox}_{\tau_k g_\theta}(x_k - \tau_k \nabla f_\theta(x_k)), \end{aligned}$$

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CONVERGENCE:

- Let $\operatorname{Argmin}_{x \in \mathbb{R}^d} f_\theta(x) + g_\theta(x) = \bigcap_{k \in \mathbb{N}} \operatorname{fix} \varphi_k$, where $\operatorname{fix} \varphi := \{x \in \mathbb{R}^N : x = \varphi(x)\}$.
- Let $f_\theta \in \Gamma_0(\mathbb{R}^N)$ convex and L -Lipschitz differentiable, $g_\theta \in \Gamma_0(\mathbb{R}^N)$. If $0 < \tau_k < 2/L$, the sequence generated by the algorithm **converge** to $\hat{x}_\theta$ [Bauschke, Combettes, 2017].
- Additionally, if f_θ is μ -strongly convex, the solution is unique and the operator φ_k turns out to be $\omega(\tau_k)$ -**Lipschitz continuous** [Taylor, Hendrickx, Glineur, 2018], where

$$(\forall \tau \in]0, 2/L[) \quad \omega(\tau) = \max\{|1 - \tau\mu|, |1 - \tau L|\} \in]0, 1[.$$

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CONVERGENCE:

- Under technical assumptions and f_θ is L -Lipschitz differentiable. If $0 < \tau_k < 1/L$, the sequence generated by the algorithm **converges** to $\hat{x}_\theta$ [Chouzenoux, Pesquet, Repetti, 2014][Bolte, Sabach, Teboulle, 2014].

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CONVERGENCE:

- Under technical assumptions and f_θ is L -Lipschitz differentiable. If $0 < \tau_k < 1/L$, the sequence generated by the algorithm **converges** to $\hat{x}_\theta$ [Chouzenoux, Pesquet, Repetti, 2014][Bolte, Sabach, Teboulle, 2014].
- Let f_θ is L -Lipschitz differentiable and g_θ is μ -weakly convex. If $0 < \tau_k < 2/(L + \mu)$, the sequence generated by the algorithm **converges** to $\hat{x}_\theta$ [Hurault, Chambolle, Leclaire, Papadakis, 2023]

Summary

Empirical perf.
Generalisation

$$\Phi_K(\cdot, \theta) = \varphi_K \circ \cdots \circ \varphi_1(\cdot, \theta)$$

$$\hat{x}_\theta = \underset{x}{\operatorname{Argmin}} f_\theta(x, z) + g_\theta(x)$$

$$\hat{x}_\theta = \lim_{K \rightarrow +\infty} \Phi_K(x_0, \theta)$$

Convex
Optimization

Robustness / Convergence

Inversion models: focus on Plug-and-Play

Minimization problem:

Iterative scheme: e.g. **Forward backward algorithm:**

$$\begin{aligned}(\forall k \in \mathbb{N}) \quad x_{k+1} &= \varphi_{k+1}(x_k, \theta) \\ &= d_{\theta}(x_k - \tau_k \nabla f_{\theta}(x_k)),\end{aligned}$$

Particular case: $x_{k+1} = d_{\theta} \left(\underbrace{x_k}_{\text{Insure prior knowledge}} - \underbrace{\gamma A^{\top} (Ax_k - z)}_{\text{Insure fidelity to forward model}} \right)$

- d_{θ} : Efficient parametric denoiser, Neural networks
- Convergence ?

Inversion models: Focus on Plug-and-Play

Minimization problem: $0 \in \nabla f_\theta + \tau^{-1}(2(\text{Id} + Q)^{-1} - \text{Id})$

Iterative scheme: e.g. **Forward backward algorithm:**

$$\begin{aligned} (\forall k \in \mathbb{N}) \quad x_{k+1} &= \varphi_{k+1}(x_k, \theta) \\ &= d_\theta(x_k - \tau_k \nabla f_\theta(x_k)), \end{aligned}$$

Particular case: $x_{k+1} = d_\theta \left(\underbrace{x_k}_{\text{Insure prior knowledge}} - \underbrace{\gamma A^\top (Ax_k - z)}_{\text{Insure fidelity to forward model}} \right)$

- If $\tau_k \equiv \tau < 2/L$ and d_θ is built to be **firmly non-expansive**. For instance $d_\theta = (\text{Id} + Q)/2$ with Q nonexpansive (1-Lipschitzian) operator, then convergence to an inclusion problem [Pesquet, Repetti, Terris, Wiaux, 2021].

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Particular case: $x_{k+1} = d_\theta \left(\underbrace{x_k}_{\text{Insure prior knowledge}} - \tau \underbrace{A^\top (Ax_k - z)}_{\text{Insure fidelity to forward model}} \right)$

- If $d_\theta = \operatorname{prox}_{g_\theta} = \operatorname{Id} - \nabla \tilde{g}_\theta$ with $\tilde{g}_\theta$ L_g -gradient Lipschitz, then convergence to a critical point if $L_f < \frac{L_g+2}{L_g+1}$ [Hurault, Chambolle, Leclaire, Papadakis, 2024]

Summary

Empirical perf.
Generalisation

$$\Phi_K(\cdot, \theta) = \varphi_K \circ \cdots \circ \varphi_1(\cdot, \theta)$$

$$\hat{x}_\theta = \underset{x}{\operatorname{Argmin}} f_\theta(x, z) + g_\theta(x)$$

$$\hat{x}_\theta = \lim_{K \rightarrow +\infty} \Phi_K(x_0, \theta)$$

PnP

Convex
Optimization

Robustness / Convergence

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Unfolded Neural Network / Unrolled proximal algorithm

→ **Operator associated to K iterations:**

$$\tilde{x}_K = \Phi_K(x_0, \theta) = \varphi_K \circ \cdots \circ \varphi_1(x_0, \theta),$$

Inversion models: Focus on unfolded/unrolled

Minimization problem: $\hat{x}_\theta \in \underset{x}{\operatorname{Argmin}} \frac{1}{2} \|Ax - z\|_2^2 + \theta \|Dx\|_1$

Analysis variational formulation

Iterative scheme: e.g. **Forward backward algorithm:**

$$u^{[k+1]} = \operatorname{prox}_{\tau\theta\|\cdot\|_1} \left(u^{[k]} - \tau DA^\top (AD^\top u^{[k]} - z) \right)$$

$$u^{[k+1]} = \underbrace{\operatorname{prox}_{\tau\theta\|\cdot\|_*}}_{\eta^{[k]}} \left(\underbrace{\operatorname{Id} - \tau DA^\top AD^\top}_{W^{[k]}} u^{[k]} + \underbrace{\tau DA^\top z}_{b^{[k]}} \right)$$

Deep learning – *General framework*

- Database : $\mathcal{S} = \{(\bar{x}_i, z_i) \in \mathbb{R}^N \times \mathbb{R}^M \mid i \in \{1, \dots, \mathbb{I}\} \text{ s.t. } z_i = A\bar{x}_i + \varepsilon\}$
- Prediction function : $d_\theta(z_i) = \eta^{[K]} (W^{[K]} \dots \eta^{[1]} (W^{[1]} z_i + b^{[1]}) \dots + b^{[K]})$

Inversion models: Focus on unfolded/unrolled

Minimization problem: $\hat{u}_\theta \in \underset{u}{\operatorname{Argmin}} \frac{1}{2} \|AD^\top u - z\|_2^2 + \theta \|u\|_1$

Synthesis variational formulation

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- Prediction function : $\operatorname{NN}_\theta(z_i) = \eta^{[K]}(W^{[K]} \dots \eta^{[1]}(W^{[1]} z_i + b^{[1]}) \dots + b^{[K]})$

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Unfolded Neural Network / Unrolled proximal algorithm

→ **Operator associated to K iterations:**

$$\tilde{u}_K = \Phi_K(u_0, \theta) = \varphi_K \circ \cdots \circ \varphi_1(u_0, \theta)$$

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Bi-level optimization problem: focus on unfolded/unrolled

Iterative scheme: e.g. **Forward backward algorithm:**

$$(\forall k \in \mathbb{N}) \quad x_{k+1} = \varphi_{k+1}(x_k, \theta)$$

Unfolded Neural Network / Unrolled proximal algorithm

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$$\tilde{x}_K = \Phi_K(x_0, \theta) = \varphi_K \circ \cdots \circ \varphi_1(x_0, \theta),$$

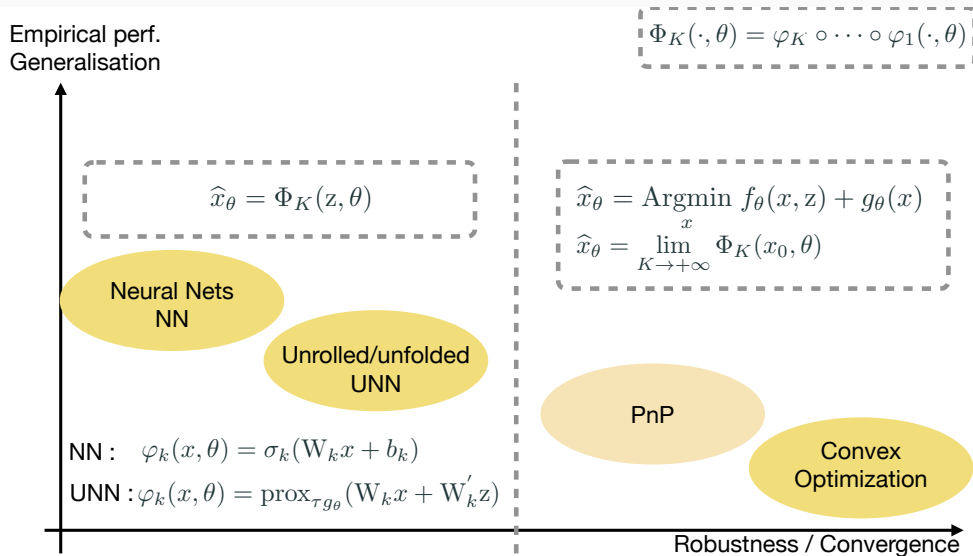
CONVERGENCE:

- Let $\hat{x}_\theta \in \bigcap_{k \in \mathbb{N}} \text{fix } \varphi_k$, where $\text{fix } \varphi(\cdot, \theta) := \{x \in \mathbb{R}^d : x = \varphi(x, \theta)\}$.
- If φ_k is ω_k -Lipschitz continuous with some $\omega_k > 0$ and $\delta_K \leq \prod_{k=1}^K \omega_k$, then

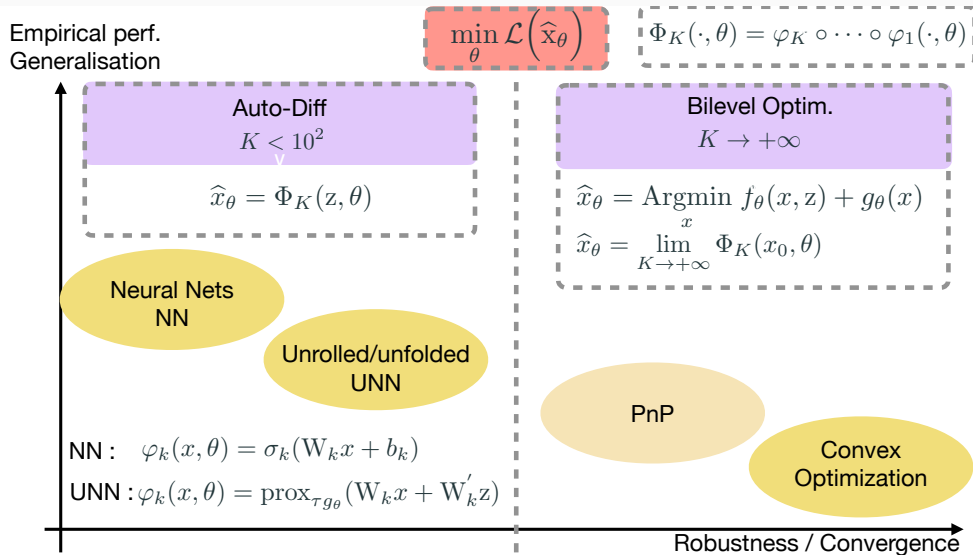
$$(\forall x_0 \in \mathbb{R}^d) \quad \|\Phi_K(x_0) - \hat{x}_\theta\| \leq \delta_K \|x_0 - \hat{x}_\theta\|.$$

- If $\omega_k \in]0, 1[$ for all k , then $\lim_{K \rightarrow \infty} \tilde{x}_K = \hat{x}_\theta$ with a linear convergence rate δ_K .

Summary



Summary



Bilevel formulation: focus on Outer problem

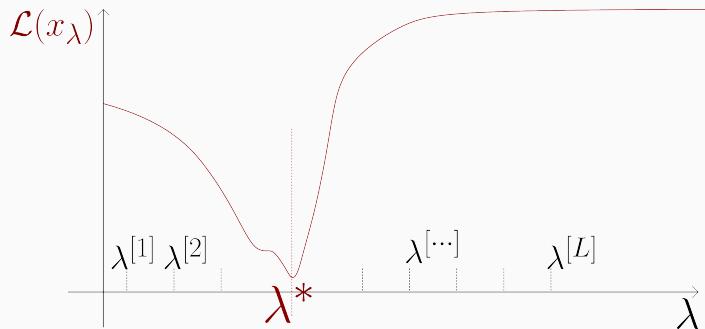
Grid-search: Given a grid $\mathcal{G} = \{\theta^{[1]}, \dots, \theta^{[N]}\}$

For $\ell = 1, 2, \dots, N$

$$\left[\tilde{x}_K^{[\ell]} = \Phi_K(z_0, \theta^{[\ell]}) \right.$$

Return $\theta^* := \theta^{[\ell^*]}$ with $\ell^* := \operatorname{argmin}_{\ell} \mathcal{L}(\tilde{x}_K^{[\ell]})$

- K large ($\hat{x}_{\theta^{[\ell]}} \approx \tilde{x}_K^{[\ell]}$)
- Difficult for $\theta \in \mathbb{R}^{P>1}$



Bilevel formulation: focus on Outer problem

AutoDiff + BackProp: Given K iterations, $\eta > 0$ a step-size :

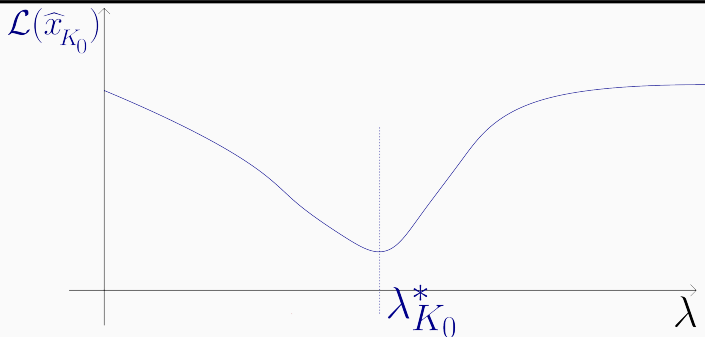
For $\ell = 1, 2, \dots$

$$\left[\begin{array}{l} \tilde{x}_K^{[\ell]} = \Phi_K(x_0, \theta^{[\ell]}) \end{array} \right.$$

Back-propagation step:

$$\left[\begin{array}{l} \theta^{[\ell+1]} = \theta^{[\ell]} - \eta \partial_{\theta} (\mathcal{L} \circ \Phi_K(x_0, \cdot))(\theta^{[\ell]}) \end{array} \right.$$

- Easy to implement
- $K \lesssim 10^2$



Bilevel formulation: focus on Outer problem

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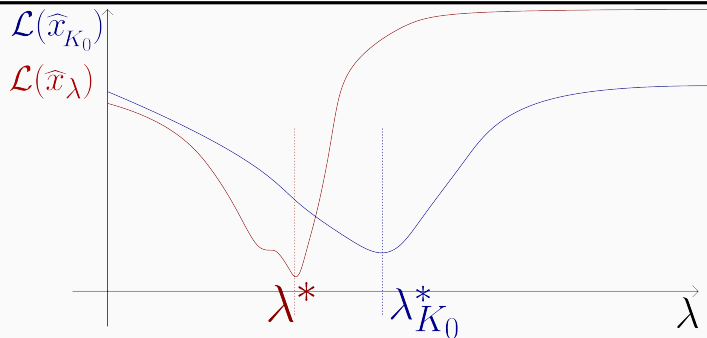
For $\ell = 1, 2, \dots$

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$$\left[\begin{array}{l} \tilde{x}_K^{[\ell]} = \Phi_K(x_0, \theta^{[\ell]}) \\ \text{Back-propagation step:} \\ \theta^{[\ell+1]} = \theta^{[\ell]} - \eta \partial_{\theta}(\mathcal{L} \circ \Phi_K(x_0, \cdot))(\theta^{[\ell]}) \end{array} \right.$$

BackProp at fixed point: Given $\eta > 0$ a step-size :

For $\ell = 1, 2, \dots$

$$\left[\theta^{[\ell+1]} = \theta^{[\ell]} - \eta \partial_{\theta}(\mathcal{L} \circ \hat{x}_{\cdot})(\theta^{[\ell]}) \right.$$

- Optimal procedure
- How to compute $\partial_{\theta} \hat{x}_{\theta^{[\ell]}}$?

Bilevel formulation: focus on Outer problem

AutoDiff + BackProp: Given K iterations, $\eta > 0$ a step-size :

For $\ell = 1, 2, \dots$

$$\left[\begin{array}{l} \tilde{x}_K^{[\ell]} = \Phi_K(x_0, \theta^{[\ell]}) \\ \text{Back-propagation step:} \\ \theta^{[\ell+1]} = \theta^{[\ell]} - \eta \partial_{\theta}(\mathcal{L} \circ \Phi_K(x_0, \cdot))(\theta^{[\ell]}) \end{array} \right.$$

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$$\left[\begin{array}{l} \theta^{[\ell+1]} = \theta^{[\ell]} - \eta \partial_{\theta}(\mathcal{L} \circ \hat{x}_{\cdot})(\theta^{[\ell]}) \end{array} \right.$$

Deep EQUilibrium (DEQ): If $\hat{x}_{\theta} = \Phi_K(\hat{x}_{\theta}, \theta)$ then

$$\partial_{\theta}(\mathcal{L} \circ \hat{x}_{\cdot})(\theta^{[\ell]}) = \partial_x \mathcal{L}(\hat{x}_{\theta^{[\ell]}}) (\mathbf{J}_{\theta}(\hat{x}_{\theta^{[\ell]}}))^{-1} \partial_{\theta} \Phi_K(\hat{x}_{\theta^{[\ell]}}, \cdot)$$

$$\text{where } \mathbf{J}_{\theta^{[\ell]}}(\hat{x}) = \text{Id} - [\partial_x(\Phi_K(\cdot, \theta^{[\ell]}))(\hat{x}_{\theta^{[\ell]}})]$$

Bilevel formulation: focus on Outer problem

Deep EQUilibrium (DEQ): Under the fixed-point identity $\hat{x}_\theta = \Phi_K(\hat{x}_\theta, \theta)$, then the hypergradient at $\hat{x}_\theta$ can be evaluated with

$$\partial_\theta \mathcal{L} \circ \hat{x}_\cdot = \partial_x \mathcal{L}(\hat{x}_\theta)^\top (\mathbf{J}_\theta(\hat{x}_\theta))^{-1} \partial_\theta \Phi_K(\hat{x}_\theta, \cdot) \quad \text{where} \quad \mathbf{J}_\theta(\hat{x}_\theta) := \text{Id} - [\partial_x(\Phi_K(\cdot, \theta))(\hat{x}_\theta)].$$

- $\mathbf{J}_\theta(\hat{x}_\theta)$ can be inverted by Neumann's series expansion [Gilton et al, 2022]:

$$(\mathbf{J}_\theta(\hat{x}_\theta))^{-1} \approx \sum_{p=0}^P [\partial_x(\Phi_K(\cdot, \theta))(\hat{x}_\theta)]^p.$$

- Jacobian-free backpropagation [Fung et al, 2022]:

$$(\mathbf{J}_\theta(\hat{x}_\theta))^{-1} = \text{Id}$$

Equivalent interpretations:

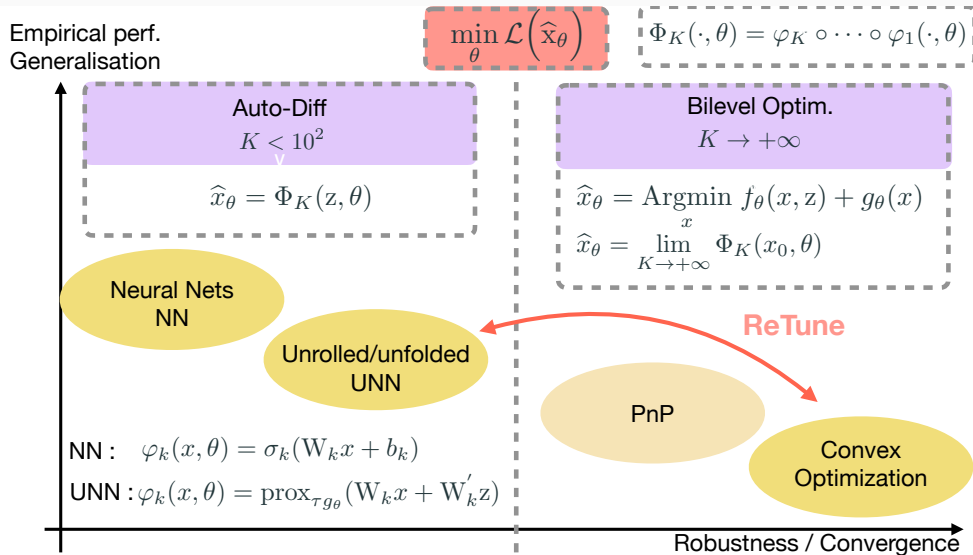
1. The Neumann inversion is truncated at $P = 0$, leading to $\mathbf{J}_\theta^{-1} = \text{Id}$.
2. From the definition of $\mathbf{J}_\theta(\hat{x}_\theta)$, it holds that $\partial_x \Phi_K(\cdot, \theta)(\hat{x}_\theta) = 0$.

Bilevel formulation: focus on Outer problem

Solving the Outer Problem relies on the gradient descent step

$$\theta^{[\ell+1]} = \theta^{[\ell]} - \tau g(\theta^{[\ell]})$$

- **At optimum:** $g(\theta) = \partial_{\theta}(\mathcal{L} \circ \hat{x})(\theta) = \partial_x \mathcal{L}(\hat{x}_{\theta})^{\top} \partial_{\theta} \hat{x}_{\theta}.$
- **DEQ:** $g^{\text{DEQ}}(\theta) = \partial_x \mathcal{L}(\hat{x}_{\theta})^{\top} J_{\theta}(\hat{x}_{\theta})^{-1} \partial_{\theta} \Phi_K(\hat{x}_{\theta}, \cdot)(\theta)$
- **Jacobian-free:** $g^{\text{JFB}}(\theta) = \partial_x \mathcal{L}(\hat{x}_{\theta})^{\top} \partial_{\theta} \Phi_K(\hat{x}_{\theta}, \cdot)(\theta)$
- **Truncated iterations :** $g^K(\theta) = \partial_x \mathcal{L}(\Phi_K(x_0, \theta))^{\top} \partial_{\theta} \Phi_K(x_0, \cdot)(\theta)$



ReTune: Restarted Truncated unrolled

T -times restarted = applying T times a contractive operator

$$(\forall x_0 \in \mathbb{R}^d) \quad x_{KT} := \Phi_K^T(x_0, \theta) := \underbrace{\Phi_K(\cdot, \theta) \circ \dots \circ \Phi_K}_{T \text{ times}}(x_0, \theta).$$

ReTune: **R**estarted **T**runcated **u**nrolled:

Set $K, T > 0$ and initialize $\theta^{[0]}$ and $x_0^{[\ell]}$.

For $\ell = 0, \dots, L$

$$\begin{cases} x_{K(T-1)}^{[\ell]} = \Phi_K^{(T-1)}(x_0^{[\ell]}, \theta^{[\ell]}) \\ \theta^{[\ell+1]} = \theta^{[\ell]} - \eta(\partial_x \mathcal{L}(x_{KT}^{[\ell]}))^{\top} \partial_{\theta} \Phi_K(x_{K(T-1)}^{[\ell]}, \cdot)(\theta^{[\ell]}) \end{cases}$$

Output: $\tilde{x}_{\theta^{[L]}, K}^T = x_{KT}^{[L]}$

Assumptions 1

For every $\theta \in \Theta$, the following hold:

1. $\Phi_K(\cdot, \theta)$ is $\delta_K(\theta)$ -Lipschitz with $\delta_K(\theta) < 1$.
2. $\{\hat{x}_\theta\} = \text{fix } \Phi_K(\cdot, \theta)$.

Theorem

Suppose that Φ_K satisfies Assumptions 1. Then, for every $\theta \in \Theta$ and $x_0 \in \mathbb{R}^n$,

$$(\forall t \in \mathbb{N}) \quad \|x_{Kt} - \hat{x}_\theta\| \leq \delta_K(\theta)^t \|x_0 - \hat{x}_\theta\|,$$

where

$$x_{Kt} := \underbrace{\Phi_K(\cdot, \theta) \circ \cdots \circ \Phi_K}_{t \text{ times}}(x_0, \theta).$$

Hence, $x_{Kt} \rightarrow \hat{x}_\theta$ as $t \rightarrow +\infty$.

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For every $\theta \in \Theta$, the following hold:

1. $\Phi_K(\cdot, \theta)$ is $\delta_K(\theta)$ -Lipschitz with $\delta_K(\theta) < 1$.
2. $\{\hat{x}_\theta\} = \text{fix } \Phi_K(\cdot, \theta)$.

Assumptions 2

For every $\theta \in \Theta$, $\partial_\theta \Phi_K(\cdot, \theta)$ is locally Lipschitz continuous at $\hat{x}_\theta$, i.e., for some $\delta > 0$, there exists $L_\theta > 0$ such that, for every $x \in \mathbb{R}^n$ satisfying $\|x - \hat{x}_\theta\|_2 \leq \delta$, we have

$$\|\partial_\theta \Phi_K(x, \theta) - \partial_\theta \Phi_K(\hat{x}_\theta, \theta)\|_2 \leq L_\theta \|x - \hat{x}_\theta\|_2.$$

Theorem If $\mathcal{L}$ is twice differentiable and Assumptions 1 and 2 are satisfied, then

$$\begin{aligned}\|g(\theta) - g^R(\theta)\|_2 &\leq \frac{\delta_K}{1 - \delta_K} \|\partial_x \mathcal{L}(\hat{x}_\theta)\|_2 \|\partial_\theta \Phi_K(\hat{x}_\theta, \cdot)\|_2 \\ &\quad + \delta_K^T L_\theta \|\hat{x}_\theta - x_0\|_2 \|\partial_x \mathcal{L}(\hat{x}_\theta)\|_2 \\ &\quad + \delta_K^T \|\hat{x}_\theta - x_0\|_2 \|\partial_{xx} \mathcal{L}(\hat{x}_\theta)\|_2 \|\partial_\theta \Phi_K(x_{K(T-1)}, \cdot)(\theta)\|_2 \\ &\quad + \|\partial_\theta \Phi_K(x_{K(T-1)}, \cdot)(\theta)\|_2 O(\delta_K^T \|\hat{x}_\theta - x_0\|_2).\end{aligned}$$

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→ Related to [Shaban, Cheng, Hatch, Boots, 2019][Bolte, Pauwels, Vaiter, 2023]

Theorem If $\mathcal{L}$ is twice differentiable and Assumptions 1 and 2 are satisfied, then

$$\begin{aligned}\|g(\theta) - g^R(\theta)\|_2 &\leq \frac{\delta_K}{1 - \delta_K} \|\partial_x \mathcal{L}(\hat{x}_\theta)\|_2 \|\partial_\theta \Phi_K(\hat{x}_\theta, \cdot)\|_2 \\ &\quad + \delta_K^T L_\theta \|\hat{x}_\theta - x_0\|_2 \|\partial_x \mathcal{L}(\hat{x}_\theta)\|_2 \\ &\quad + \delta_K^T \|\hat{x}_\theta - x_0\|_2 \|\partial_{xx} \mathcal{L}(\hat{x}_\theta)\|_2 \|\partial_\theta \Phi_K(x_{K(T-1)}, \cdot)(\theta)\|_2 \\ &\quad + \|\partial_\theta \Phi_K(x_{K(T-1)}, \cdot)(\theta)\|_2 O(\delta_K^T \|\hat{x}_\theta - x_0\|_2).\end{aligned}$$

Corollary Let $\bar{x} \in \mathbb{R}^N$ and set $\mathcal{L}(x) = \frac{1}{2} \|x - \bar{x}\|^2$, then

$$\begin{aligned}\|g(\theta) - g^R(\theta)\|_2 &\leq \frac{\delta_K}{1 - \delta_K} \|\hat{x}_\theta - \bar{x}\|_2 \|\partial_\theta \Phi_K(\hat{x}_\theta, \cdot)\|_2 \\ &\quad + \delta_K^T \|x_0 - \hat{x}_\theta\|_2 (\|\partial_\theta \Phi_K(x_{K(T-1)}, \cdot)(\theta)\|_2 + L_\theta \|\hat{x}_\theta - \bar{x}\|_2).\end{aligned}$$

Numerical experiments: Wavelet denoising with $z = \bar{x} + \varepsilon$

Minimization problem: $\hat{x}_\theta = \operatorname{argmin}_x \frac{1}{2} \|x - z\|_2^2 + \|Dx\|_{1,2,\theta}$

where

1. $\|\cdot\|_{1,2,\theta}$ is a parametric prior imposing regularity on wavelet coefficients.
2. $\ell_{2,1}$ -norm with weighted bands and channels:

$$(\forall w \in \mathbb{R}^N) \quad \|w\|_{1,2,\theta_{B,C}} := \sum_j \lambda_j \sum_k \sqrt{\sum_{b,c} \Lambda_{b,c} w_{j,k,b,c}^2};$$

3. $\ell_{2,1}$ -norm with weighted bands:

$$(\forall w \in \mathbb{R}^N) \quad \|w\|_{1,2,\theta_B} := \sum_j \lambda_j \sum_{k,c} \sqrt{\sum_b \Lambda_b w_{j,k,b,c}^2}.$$

Numerical experiments: Wavelet denoising with $z = \bar{x} + \varepsilon$

Minimization problem: $\hat{x}_\theta = \operatorname{argmin}_x \frac{1}{2} \|x - z\|_2^2 + \|Dx\|_{1,2,\theta}$

Reformulation $\hat{u}_\theta = D^\top \theta^{-1} \operatorname{argmin}_u \frac{1}{2} \|D^\top \theta^{-1} u - z\|_2^2 + \|u\|_{1,2}.$

Iterative scheme: e.g. **Forward backward algorithm:**

$$u_{k+1} = \operatorname{prox}_{\gamma\theta\|\cdot\|_1} \left(u_k - \gamma D A^\top (A D^\top u_k - z) \right) = \varphi(u_k, \theta)$$

Numerical experiments: Wavelet denoising with $z = \bar{x} + \varepsilon$

Minimization problem: $\hat{x}_\theta = \operatorname{argmin}_x \frac{1}{2} \|x - z\|_2^2 + \|Dx\|_{1,2,\theta}$

Iterative scheme: e.g. **Forward backward algorithm:**

$$u_{k+1} = \operatorname{prox}_{\gamma\theta\|\cdot\|_1} \left(u_k - \gamma DA^\top (AD^\top u_k - z) \right) = \varphi(u_k, \theta)$$

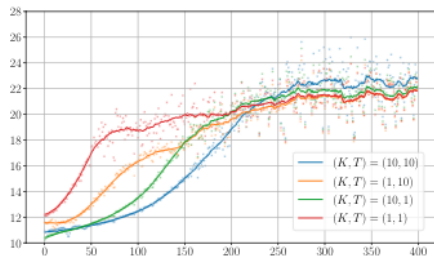
→ **Operator associated to K iterations:**

$$\tilde{u}_K = \Phi_K(u_0, \theta) = \varphi_K \circ \cdots \circ \varphi_1(u_0, \theta)$$

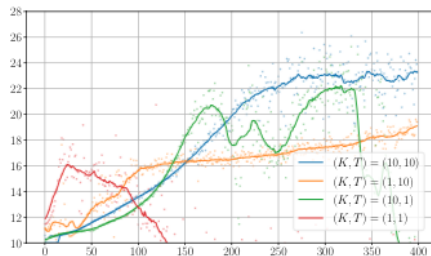
→ **T -times restarted**

$$(\forall x_0 \in \mathbb{R}^d) \quad u_{KT} := \Phi_K^T(u_0, \theta) := \underbrace{\Phi_K(\cdot, \theta) \circ \cdots \circ \Phi_K}_{T \text{ times}}(u_0, \theta).$$

Numerical experiments: Wavelet denoising with $z = \bar{x} + \varepsilon$

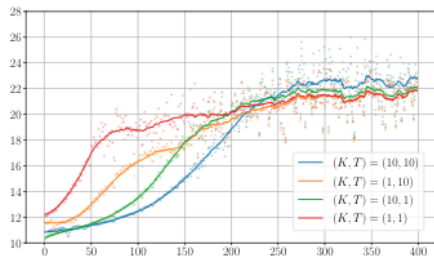


(a) Evolution of PSNR evaluated on the training dataset to learn θ_B

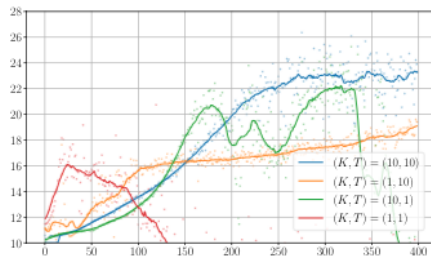


(b) Evolution of PSNR evaluated on the training dataset to learn $\theta_{B,C}$

Numerical experiments: Wavelet denoising with $z = \bar{x} + \varepsilon$

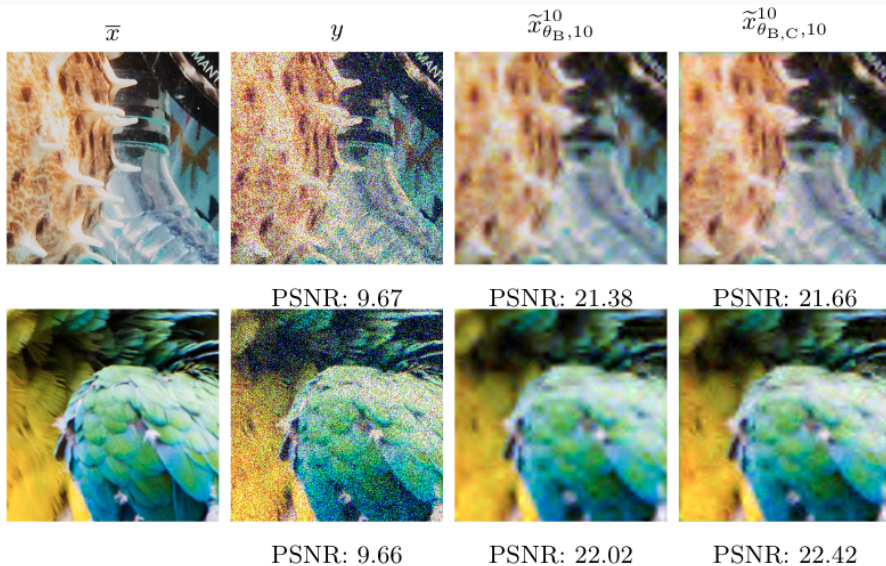


(a) Evolution of PSNR evaluated on the training dataset to learn θ_B



(b) Evolution of PSNR evaluated on the training dataset to learn $\theta_{B,C}$

Numerical experiments: Wavelet denoising with $z = \bar{x} + \varepsilon$



Numerical experiments: Image inpainting with PnP $z = A\bar{x} + \varepsilon$

Minimization problem:

Iterative scheme: e.g. **Forward backward algorithm:**

$$\begin{aligned}x_{k+1} &= D_{\Lambda, \sigma}(x_k - \tau A^\top (Ax_k - z)) \\ &= \varphi(x_k, \theta)\end{aligned}$$

with learnable parameters

- τ a stepsize
- D_Λ a Drunet denoiser with parameters (filter, bias, ...) Λ
- σ the noise level of the denoiser

Numerical experiments: Image inpainting with PnP $z = A\bar{x} + \varepsilon$

Minimization problem:

Iterative scheme: e.g. **Forward backward algorithm:**

$$\begin{aligned}x_{k+1} &= D_{\Lambda, \sigma}(x_k - \tau A^\top (Ax_k - z)) \\ &= \varphi(x_k, \theta)\end{aligned}$$

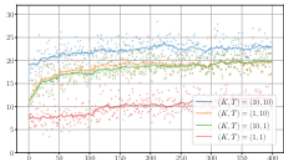
→ **Operator associated to K iterations:**

$$\tilde{u}_K = \Phi_K(u_0, \theta) = \varphi_K \circ \cdots \circ \varphi_1(u_0, \theta)$$

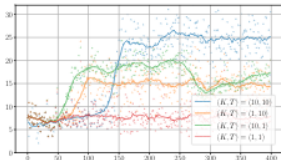
→ **T -times restarted**

$$(\forall x_0 \in \mathbb{R}^d) \quad x_{KT} := \Phi_K^T(x_0, \theta) := \underbrace{\Phi_K(\cdot, \theta) \circ \cdots \circ \Phi_K}_{T \text{ times}}(x_0, \theta).$$

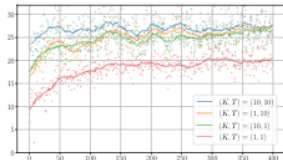
Numerical experiments: Image inpainting with PnP $z = A\bar{x} + \varepsilon$



(a) Evolution of PSNR evaluated on the training dataset to learn θ_B in the FB scheme

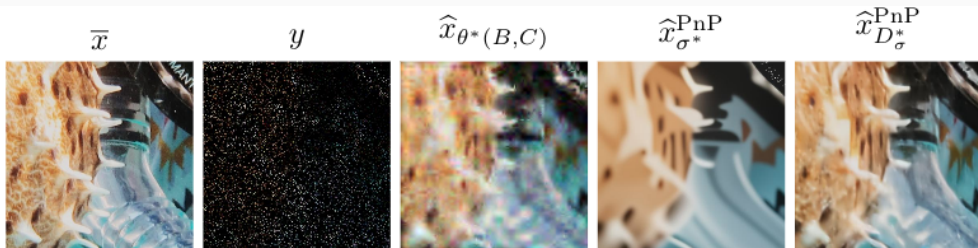


(b) Evolution of PSNR evaluated on the training dataset to learn $\theta = (\sigma, \tau)$ in the FB-PnP scheme



(c) Evolution of PSNR evaluated on the training dataset to learn $\theta = \Lambda$ in the FB-PnP scheme

Experiments: Inpainting reconstruction results with ReTune



PSNR: 5.06

PSNR: 20.96

PSNR: 21.16

PSNR: 23.00



PSNR: 6.07

PSNR: 21.42

PSNR: 22.10

PSNR: 23.91

Conclusions and perspectives

- Deep learning brings important improvements and new theoretical challenges in image restoration.
- ReTune: Train truncated unrolled algorithm using automatic differentiation techniques with theoretical guarantees.
- ReTune: Truncated neural network can be restarted to infer solutions of

$$\hat{x}_\theta \in \underset{x}{\operatorname{Argmin}} f_\theta(x) + g_\theta(x)$$

- ReTune: way to benefit of theoretical guarantees of PnP with performance of unfolded/unrolled methods.

Conclusions and perspectives

- ReTune: procedure to solve bilevel formulation in a simple fashion, with learning steps that can be made arbitrarily close to the optimal DEQ steps.
- Lipschitz properties of Forward-Backward iterations were leveraged to show the interplay between the depth and the number of restarts in **ReTune**.
 - depth K : controls the approximation error of a Jacobian-free step,
 - restart T : unrolled neural network allows one to reach the equilibrium point.
- ReTune on DeepInverse
- Future work: Deeper analysis of ReTune for relaxed assumptions.

L. Davy, LM Briceno-Arias, N. Pustelnik, Restarted contractive operators to learn at equilibrium, HNA 26: Machine Learning Solutions for Inverse Problems: Part A, 2025.

L. Davy, P. Abry, N. Pustelnik, Restart strategies enabling automatic differentiation for hyperparameter tuning in inverse problems, EUSIPCO Lyon, 2024.